Digital Systems

Engineering Technology

May, 2026

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Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Digital Systems (A11922)

Short Title:	Digital Systems		
Faculty	Science and Computing		
Department	Engineering Technology		
Cluster	Electronics		
Author	CLAIRE.FITZPATRICK		
Credits	5	**Level:**	Intermediate

Description of Module / Aims

This module follows on from the module called Finite State Machines and builds on concepts introduced there. As the required size and complexity of digital systems increases, the designer is forced to implement the design using specialised devices,programmable logic devices(PLDs') and field-programmable gate arrays(FPGAs'). This module will study the options available and examine HDL in terms of its translation into hardware elements or synthesis. The operating characteristics, manufacturers specifications and applications of D/A and A/D converters will be investigated. Address decoding schemes, together with good design principles and troubleshooting techniques will also be covered.

Programmes

`DSYS-0001`	BEng (Hons) in Electronic Engineering (`WD_EONIC_B`)	2 / 4 / M
	BEng (Hons) in Information Engineering (International) (`WD_EEELC_BI`)	3 / 5 / M

Indicative Content

- Memory Devices
- Programmable Logic Devices
- Address Decoding Schemes
- A/D and D/A Conversion
- Synthesis using HDL
- Good Design Principles
- Troubleshooting Digital Systems

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Explain how ROM and RAM devices operate, how they may be expanded to increase word length and capacity and incorporated in address decoding schemes.
2. Appreciate the flexibility of programmable logic devices and undertake synchronous and asynchronous design solutions using PLDdevices.
3. Use HDL as a basis for FSM synthesis.
4. Analyse D/A and A/D conversion techniques, interpret the relevant performance characteristics and recognise typical conversion errors.
5. Demonstrate the importance of good design principles and the significance of troubleshooting techniques.
6. Construct, test and simulate specified digital systems, relevant to the theory lectures in a laboratory setting.

Learning and Teaching Methods

- Lectures

- Practicals /Project
- Integrated demonstrations and examples

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	1,2,3,4,5
Continuous Assessment	50%	
In-Class Assessment	15%	1,2,3,4,5
Practical	35%	3,5,6

Assessment Criteria

70% - 100%: The learner has: attained the module learning outcomes at an excellent level; demonstrated a comprehensive knowledge of the associated subject matter; achieved an excellent level of the skills required for the subject matter; demonstrated the ability to carry out further investigation and problem solving associated with the subject matter.

60% - 69%: The learner has: attained the module learning outcomes at a very good level; demonstrated a detailed knowledge of the associated subject matter; achieved a very good level of the skills required for the subject matter.

50% - 59%: The learner has: attained the module learning outcomes at a good level; demonstrated a good knowledge of the associated subject matter; achieved a good level of the skills required for the subject matter.

40% - 49%: The learner has: attained the module learning outcomes at a basic level; demonstrated a basic knowledge of the associated subject matter; achieved a basic level of the skills required for the subject matter.

<40%: The learner has not: attained the module learning outcomes; demonstrated sufficient knowledge of the associated subject matter; demonstrated a sufficient level of the skills required for the subject matter.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Practical	12	
Independent Learning	87	

Essential Material

- Floyd, T. _Digital Fundamentals_. 11th ed.. New Jersey: Pearson Education, 2015.
- Lewin, D. and D. Protheroe. _Design of Logic Systems_. 2nd ed.. London: Chapman & Hall, 1992.
- Wakerly, J. _Digital Design, Principles and Practices_. 4th ed.. New Jersey: Prentice Hall, 2001.

Supplementary Material

- "Orcad." www.orcad.com
- "Xilinx." www.xilinx.com
- Ashenden, P. _The Students Guide to VHDL_. San Francisco: Morgan Kauffman, 2008.

- Katz, H. and G. Borriello. _Contemporary Logic Design_. 2nd ed.. New Jersey: Prentice Hall, 2005.
- Palnitkar, S. _Verilog HDL_. 2nd ed. New Jersey: Prentice-Hall, 2003.

Requested Resources

- ENGINEERING LAB: Electronics